Covid-19 Australia Prediction

Project Report

DS424: Time Series and Forecasting for Data Science

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Introduction

Forecasting time series has now become an important analytical technique in modern-day data science, as it enables the researcher and decision-maker to understand the change of events over time and to forecast future changes with increased accuracy. Many applications in the real world require modeling of a temporal pattern, which, in turn, is often not just desirable but an essential requirement for early intervention and informed policymaking. This project studies the progress of COVID-19 new deaths in Australia and builds a time-series forecasting model that can grasp the monthly trend and predict future outcomes. Understanding how health indicators change over time is very important for the development of strategies that reduce risks and enhance preparedness. Time series forecasting-which utilizes the examination of historical patterns and modeling of future behavior-represents a very powerful method of translating data into insights of true meaning. This current project uses statistical forecasting methods to show how data-driven analyses inform effective planning, early warning, and resource allocation with respect to global health events such as the COVID-19 pandemic.

Besides its technical contribution, this work is in line with one of the United Nations SDGs, namely SDG 3 - Good Health and Well-being - whose core is to enhance the capacity of all countries for early warning, risk reduction, and management of national and global health risks. The forecasting of COVID-19 trends in this project contributes to that vision by showing how the use of analytics tools best fosters improved health monitoring and sustainable evidence-based decision making.

1. Preparing Data

The objective of this chapter was to clean the dataset and put it into a structure that will make it suitable for time series analysis. We called R libraries required: **tidyverse**, **tsibble**, and **lubridate**, as these utilities make it much easier to clean the data, work with dates, and create structures suitable for time-based analysis. To avoid any issues with file paths, we imported the dataset manually via a simple file selection window. We then loaded the OWID COVID-19 dataset, focusing only on data relating to **Australia**, and selected columns most relevant to our purposes-**location**, **date**, and **new_deaths**. These variables allowed us to easily track how new COVID-19 deaths changed over time. A **year-month** column was then created using the **yearmonth()** function, which helped convert the daily data into a much clearer monthly layout. We further dealt with missing values by replacing empty or undefined values in **new_deaths** with zero. After that, we grouped the data by **location** and month and calculated the total number of **new_deaths** for each month, producing a clean and consistent monthly summary of the data. Lastly, we transformed monthly-level data to a **tsibble**, which is how the data should be structured for time series analysis in R. To make sure all were processed correctly, we showed the first ten rows of the final dataset [1] [2] [3].

Filter and select necessary columns: Filter and select necessary columns:

```
aus <- covid %>%
  filter(location == "Australia") %>%
  select(location, date, new_deaths)
#-------------------
```

Figure 1.1: Filter and select code

Handle missing values (with the output), Summarize and group by necessary columns:

```
> # Handle missing values before aggregation and handing it
> aus <- aus %>%
+   mutate(new_deaths = if_else(is.na(new_deaths), 0, new_deaths))
> sum(is.na(aus$new_deaths))
[1] 0
```

```
# Aggregate monthly new deaths
monthly_data <- aus %>%
  group_by(location, month) %>%
  summarize(
    new_deaths = sum(new_deaths, na.rm = TRUE)
  ) %>%
  ungroup()
```

Figure 1.2: Handling Missing Values (with the output), Summarize code

Convert the data into a tsibble object :

```
# Convert the monthly data into a tsibble object
monthly_tsibble <- monthly_data %>%
  as_tsibble(index = month, key = location)
#-------------------
```

Figure 1.3: Conversion code

First 10 rows:

```
> monthly_tsibble %>% head(10)
# A tsibble: 10 x 3 [1M]
# Key:       location [1]
   location     month new_deaths
   <chr>        <mth>      <dbl>
 1 Australia 2020 Jan          0
 2 Australia 2020 Feb          0
 3 Australia 2020 Mar         16
 4 Australia 2020 Apr         69
 5 Australia 2020 May         22
 6 Australia 2020 Jun          1
 7 Australia 2020 Jul         66
 8 Australia 2020 Aug        550
 9 Australia 2020 Sep        163
10 Australia 2020 Oct         27
```

Figure 1.4: Displaying first 10 rows

2. Data Visualization

Data visualization helps present the monthly new COVID-19 deaths in a clearer way using different graphical tools. These visuals make it easier to inspect trend, seasonality, cyclic movement, autocorrelation, and to check whether the data behaves like white noise. Such techniques support understanding the time series and preparing for model selection [4].

2.1 Time Plot (Autoplot)

The line graph shows COVID-19 deaths in Australia by month to demonstrate how the data pattern evolved throughout time. The death numbers remained at a low level throughout 2020 before reaching high numbers during the first 2022 outbreak which then decreased steadily until 2023 and 2024. The data shows an increasing pattern instead of periodic fluctuations because the value increases occur irregularly. The graph shows no seasonal patterns because its changes stem directly from pandemic outbreaks instead of following monthly patterns. The graph shows pandemic spread patterns across different time periods but it does not show any repeating patterns or seasonal trends.

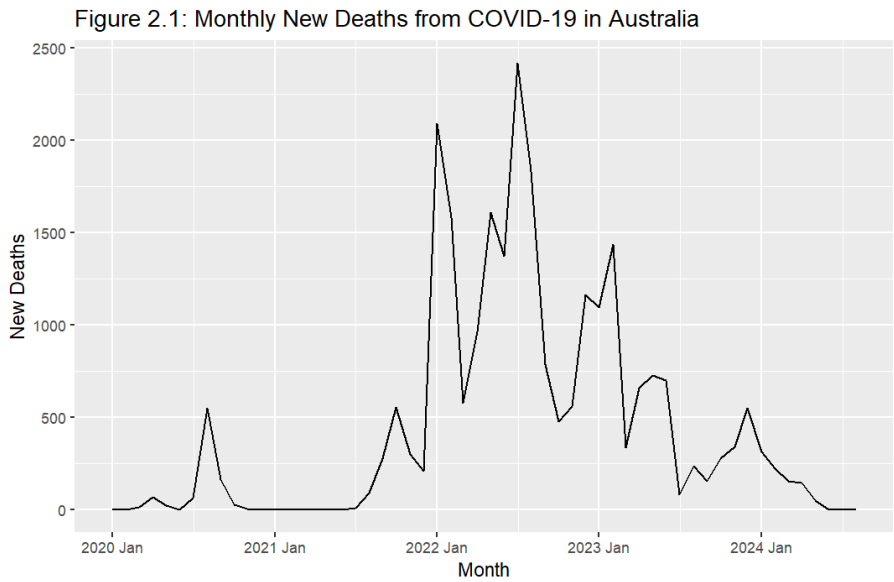

Figure 2.1: Autoplot of Monthly COVID-19 Deaths in Austalia

2.2 Seasonal Plot

The subseries plot helps show how each month behaves across different years. Instead of looking at the entire time series as one line, this plot groups all values for the same month together. This makes it easier to notice whether certain months usually have higher or lower numbers of new COVID-19 deaths. In our data, the monthly groups do not follow a clear or repeated seasonal pattern. Some months are high in certain years and low in others, which suggests that the changes are mainly caused by irregular COVID waves rather than a stable seasonal effect.

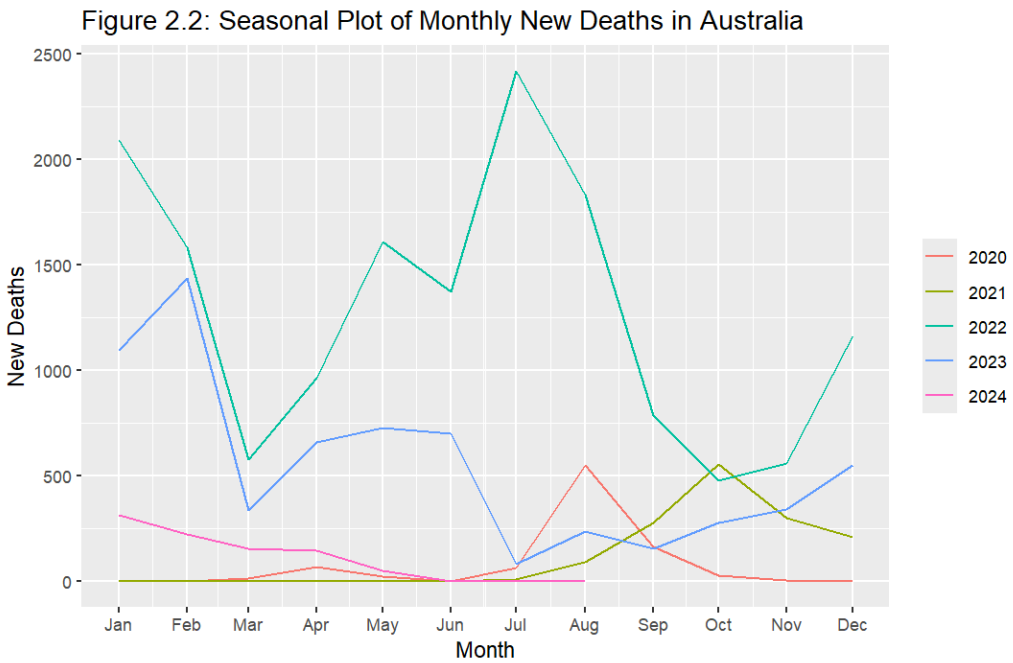

Figure 2.2 : Seasonal plot of Monthly new Deaths in Austalia

2.3 Subseries Plot

The subseries plot demonstrates how different months perform throughout various years. The plot enables viewers to identify which months tend to record more or fewer new COVID-19 deaths. The plot organizes all monthly values into separate groups instead of displaying them as a single continuous line. The monthly data points in our dataset show no evidence of seasonal patterns throughout the months. The data indicates that COVID-19 death rates during different months between years follow unpredictable patterns instead of seasonal fluctuations.

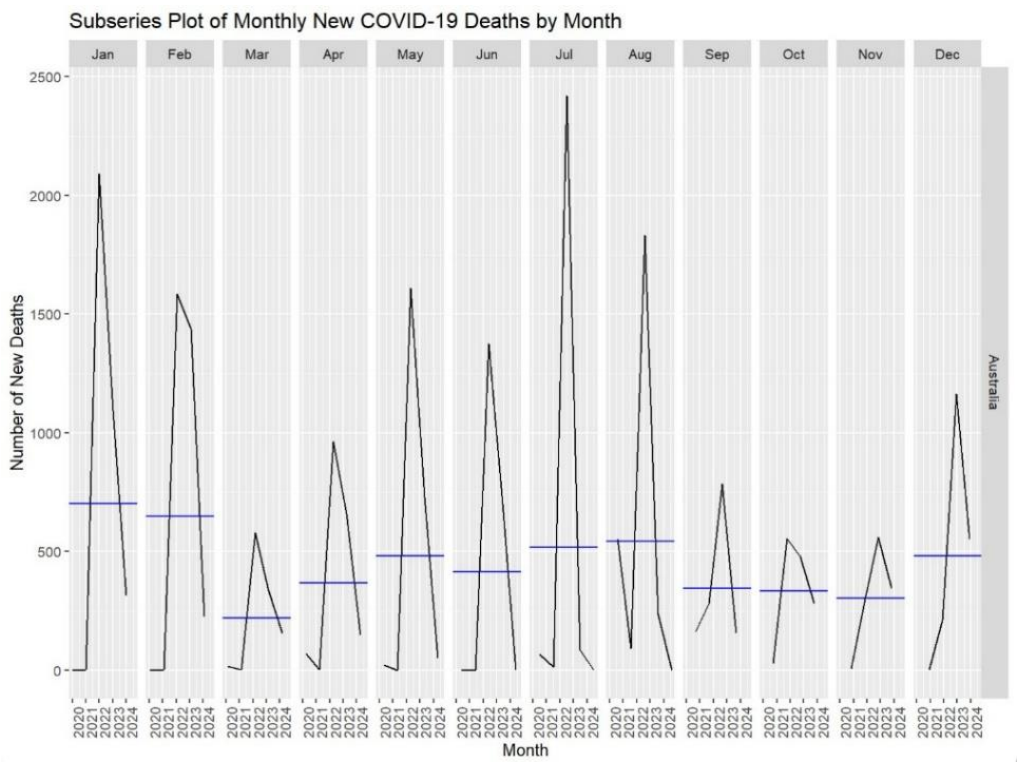

Figure 2.3 : Subseries plot of Monthly new COVID-19 Deaths by Month

2.4 Autocorrelation Function (ACF)

The autocorrelation plot in Figure 2.4 shows how each month relates to the months before it. The first few lags have clear positive autocorrelation, and several points fall outside the significance limits. This tells me that the number of new deaths in a given month is strongly connected to earlier months. This pattern is common in epidemic data because changes usually develop over time. Since the series shows real dependence and not random movement, it is suitable for forecasting models that use past values to predict future ones.

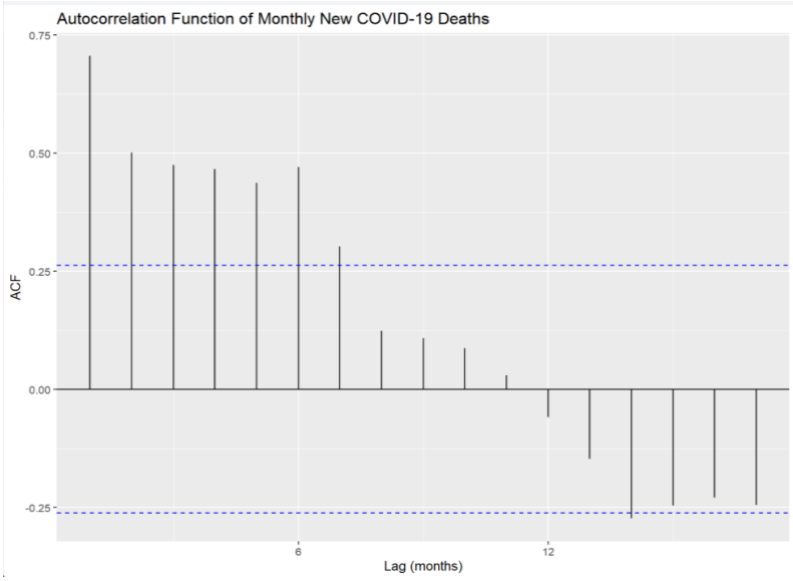

Figure 2.4: Autocorrelation Function of Monthly new COVID-19 Deaths

2.5 Decomposition of the Time Series

The STL decomposition helps break the series into three parts so I can understand its behavior more clearly. Figure 2.5 shows the original data along with the trend, seasonal pattern, and remainder. The trend line shows how the number of new deaths changes over time and makes the main waves easier to see. The seasonal part is weak and not consistent across the years, which means the series doesn't follow a strong yearly pattern. The remainder still has a few spikes, and these are mostly linked to sudden shifts in the pandemic. Overall, the decomposition shows that the series has a clear trend and some cyclic movements, but the seasonality is irregular. This helps guide the choice of forecasting models later, especially models like ARIMA that can handle non-stationary data with time dependence [5].

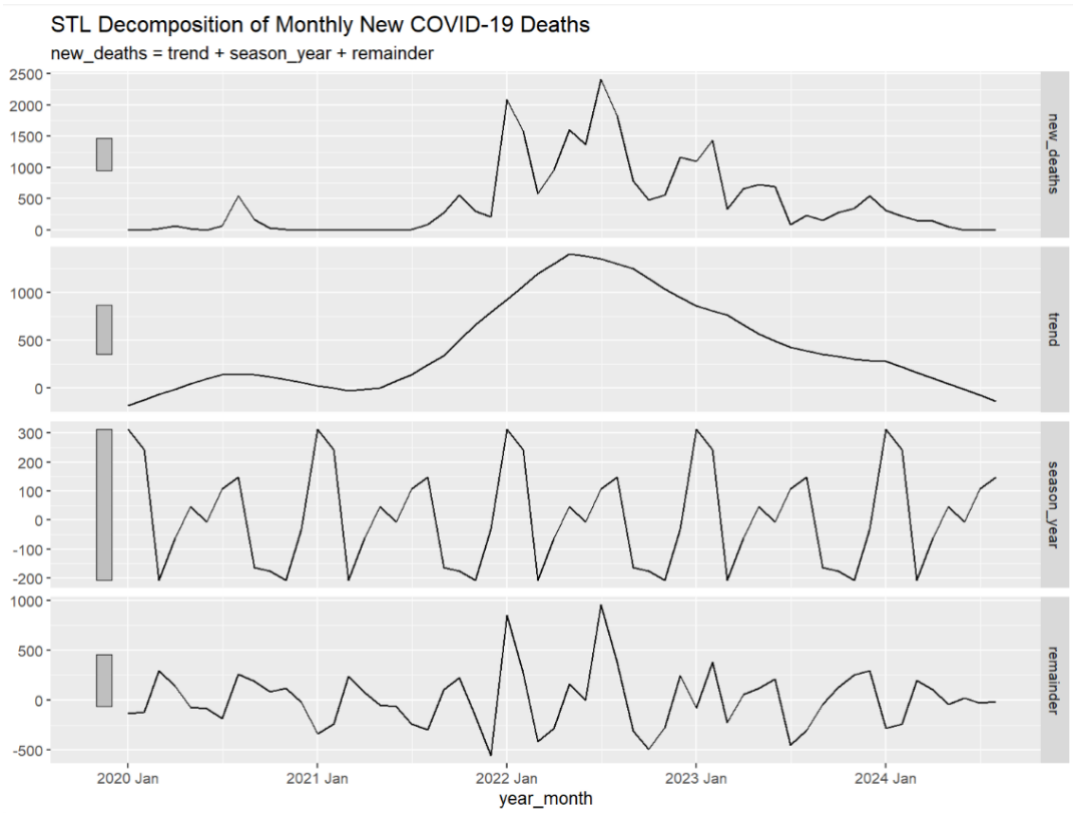

Figure 2.5: STL Decomposition of Monthly new COVID-19 Deaths

3. Specifying Model

The exploratory analysis shows that COVID-19 new-deaths data shows a moving pattern with multiple peaks, but seasonal and subseries plots show no evidence of recurring seasonal patterns. The ACF results show that the data contains strong autocorrelation during its initial lags, but STL decomposition shows that trend and irregular components explain most of the data variation instead of seasonal components. The data shows non-stationary patterns because trends dominate all other components and there are no detectable seasonal patterns. The data follows specific patterns which make it suitable for modeling through four forecasting methods [5] [6]:

3.1 Holt's Linear Exponential Smoothing

The series shows a changing pattern and trend but no dependable seasonal pattern so Holt's linear method (ETS(A,A,N)) represents an appropriate model selection. The model design works best for non-seasonal data that shows changing trends, and it successfully handles time-based series with signal changes which are common in epidemic data [7].

3.2 Damped Trend Exponential Smoothing

The series demonstrates shifting patterns through various changes while its trend patterns become less pronounced throughout the time span. The observed pattern in the data supports the implementation of the ETS(A,Ad,N) damped trend model because it extends Holt's linear method to produce declining trends instead of endless growth. The model uses the damping parameter to represent the increasing and decreasing movements observed in the COVID-19 data. The damped trend model shows excellent performance for non-seasonal epidemic data because it shows trend development while trend power decreases with time [7].

3.3 Non-seasonal ARIMA

The ACF pattern and STL decomposition trend identification require differencing to achieve stationarity which makes ARIMA(p,1,q) the most suitable model. The data lacks stable seasonal patterns so including seasonal ARIMA terms becomes unnecessary. The early significant autocorrelation values indicate that both autoregressive and moving-average terms should be included in the model while information-criterion-based automated selection will determine the optimal parameter values [8] [5].

3.4 Prophet

The monthly COVID-19 new-deaths data pattern makes Prophet an appropriate modeling solution for this dataset. The exploratory analysis confirms that these models represent the best options for working with this specific dataset [6]. The four selected models match the characteristics of the monthly COVID-19 new-deaths time series because it exhibits changing patterns and lacks seasonal consistency and displays non-linear wave-like patterns. [6].

The four selected models fit the characteristics of the monthly COVID-19 new-deaths time series, which shows shifting trends [9], no stable seasonal pattern, and non-linear wave-like behaviour. The exploratory analysis indicates that these models are the most suitable choices for this dataset [6].

4. Model Estimation

This chapter presents the estimation of the time-series models using the training dataset, defined as a result of splitting the original series into a training and testing subset. The former includes all observations before August 2022, while the latter encompasses all the data afterwards. Such a temporal split of data is viewed as a standard way of treating time-series forecasting problems because it enforces model estimation to rely exclusively on past information and avoids data leakage threatening chronological integrity [8].

The estimation stage involved fitting four forecasting models: Holt's Linear Exponential Smoothing, Damped Trend Exponential Smoothing, Non-seasonal ARIMA, and the Prophet model. The first three models have been estimated using the fable framework; each of these models has been specified within the model() function and directly applied to the training dataset. This resulted in model objects containing the estimated parameters, level and trend components, along with other structural characteristics integral to understanding the fitted behaviour of each model [8].

The data was first reformatted for the Prophet model by renaming the date column to ds and the target variable to y, as structured in the requirements of Prophet. Estimating the model utilized the function prophet(), which auto-detects trend shifts, changepoints, and monthly patterns [8][9].

The outputs from the estimation process for each model individually basically confirm that estimation was successfully achieved, and this means that all models are ready for further steps of the analysis. Chapter 5 proceeds with the evaluation and comparisons of their performances using accuracy metrics, whereas Chapter 6 uses the fitted models to produce future forecasts [8].

5. Model Evaluation

The hold-out evaluation was used to compare how well each model predicts the testing period after August 2022. The results show clear differences in performance across the models. Holt's Linear and the Damped Trend model produced relatively large errors because they could not fully capture the sharp changes in COVID-19 death counts. The ARIMA model performed better than both smoothing models and showed smaller RMSE and MAE values, which indicates that it handled the data pattern more effectively [10]. Prophet achieved the lowest error values for several months in the testing period and was able to adapt to the changes in the trend more smoothly. Since some months in the test set have zero deaths, MAPE was not reliable, so the evaluation mainly depended on MAE and RMSE. Based on these metrics, Prophet and ARIMA provided the strongest performance in the hold-out test, with Prophet showing more stable month-to-month predictions.

Model	ME	MAE	RMSE	MAPE
Holt's Linear	-4145	4145	4549	Inf
Damped Trend	-2535	2535	2684	Inf
Non-seasonal ARIMA	-2201	2201	2368	Inf
Prophet	-1449	1589	1847	Inf
Bagged ARIMA (Bootstrap)	-29.2	361	465	Inf

several months in the testing period have zero new deaths, the MAPE values become infinite and are therefore not reliable. For this reason, the comparison mainly focuses on MAE and RMSE.

5.1 Cross-Validation

To check how consistent each model is across different time periods, rolling time series cross-validation was applied. This method tests the models repeatedly using several training windows and short-term forecasts. The results showed that the Damped Trend and Holt's Linear models achieved similar accuracy, with both producing moderate errors across the folds. ARIMA performed slightly worse in cross-validation because its performance varied more across different time windows [11], which reflects the strong irregular movements in the data. Overall, the cross-validation results support the idea that no single model performs perfectly in every

time period, but the smoothing models remain more stable while Prophet and ARIMA continue to show stronger performance during periods with major trend changes.

5.2 Bootstrapping

Bootstrapping and bagging were used as an additional method to improve forecast stability. Several bootstrap samples were created from the training data [12], and an ARIMA model was fitted to each sample. The final forecast was computed by taking the average of all bootstrap predictions, which reduces noise and smooths out large fluctuations. The bagged model achieved noticeably lower MAE and RMSE compared to the individual ETS and smoothing models, and it performed better than the regular ARIMA model on the testing set. This shows that combining multiple bootstrap forecasts can provide a more stable and reliable prediction, especially for data such as COVID-19 deaths where sudden changes are common.

6. Producing Forecast

This chapter presents the final forecasts generated from the models estimated earlier Holt's Linear, Damped Trend, Non-seasonal ARIMA, Prophet, and the Bagged ARIMA model. The forecasts cover the 12 months following August 2022 and illustrate how each model interprets and projects the future pattern of COVID-19 mortality in Australia.

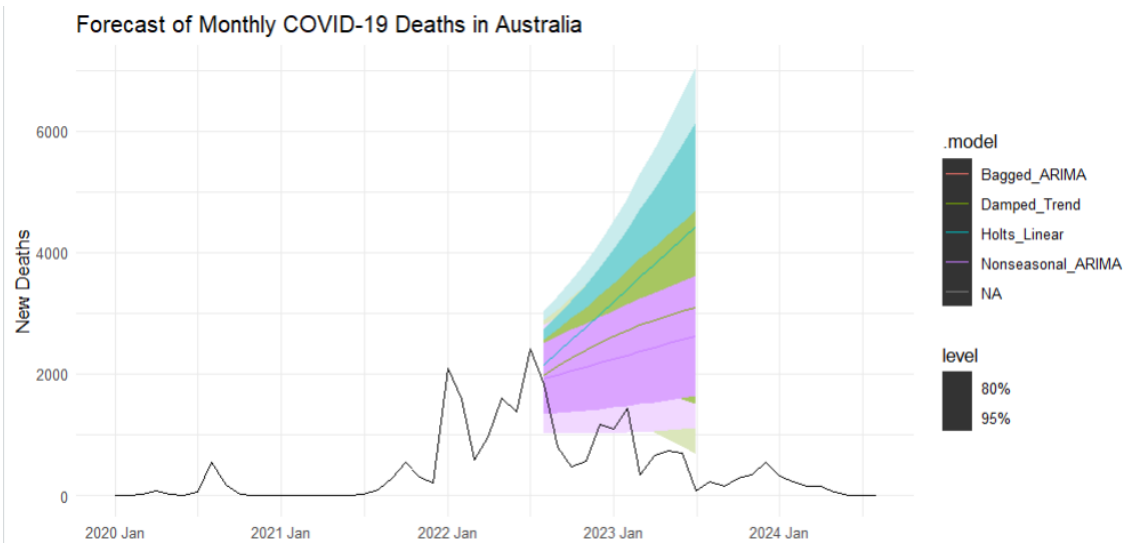

Figure 6.1: Forecast models

The forecast plot shows a clear divergence in how the models behave. Holt's Linear and the Damped Trend models produce rapidly increasing forecasts with wide prediction intervals. This makes sense because smoothing models overreact to sudden spikes. Since the data contains sharp jumps, the forecasts naturally become unstable, causing these methods to extrapolate short-lived peaks into unrealistic long-term growth. This behavior is common when smoothing models are applied to epidemic data characterized by structural shocks and abrupt, irregular movements.

The Non-seasonal ARIMA model generates forecasts that are less extreme than the ETS models yet still show substantial unpredictability. The model remains highly sensitive to the strong non-stationarity and abrupt changes that characterize COVID-19 mortality data. Although it captures overall trend direction more accurately, it loses stability when predicting future values beyond the short term because the data exhibits irregular and non-linear patterns. Prophet provides comparatively smoother and more stable forecasts. Its ability to automatically detect changepoints and adjust to non-linear trend variations prevents the explosive behavior seen in the ETS models. Prophet aligns well with the wave-like structure of the data and produces narrower forecast intervals with more realistic projections.

The Bagged ARIMA model delivers the most stable and conservative forecasts. By averaging predictions across multiple bootstrap samples, it reduces variance and minimizes the influence of extreme observations. This approach helps counteract the large spikes observed in 2022, resulting in more reliable trajectories than those produced by individual models. The forecasting results show that although the selected models follow the statistical patterns found in the exploratory analysis, the unpredictability of epidemic data limits their ability to produce accurate long-term forecasts [13]. The fast-changing COVID-19 death statistics in Australia make predicting future values difficult because the models generate highly uncertain results. Prophet and Bagged ARIMA produce the most accurate outcomes because they detect sudden changes and improve forecast performance through ensemble averaging. The results highlight that statistical models require epidemiological understanding, as pandemic data volatility

makes forecasting methods less reliable. The generated forecasts demonstrate how the models perform and support the main analytical findings of this project.

The creation of COVID-19 mortality trend prediction models helps achieve SDG 3 by creating improved early warning systems which enhance public health readiness during rapid changes in mortality rates [14]. The research under SDG 9 shows how data innovation improves national forecasting systems through Prophet and ARIMA and bootstrapped ensemble models [15]. The project achieves SDG 17 through its implementation of international open-source datasets and standardized modeling practices which enable global data sharing and analytical method standardization between countries. The project shows that statistical modeling of shared global data resources makes forecasting more effective which leads to better health system stability and stronger data infrastructure and international research partnerships [16].

In summary, the forecasting results show that epidemic time-series remain highly difficult to predict due to their irregular and fast-changing patterns. While traditional smoothing models produced unstable long-term projections, more advanced approaches such as Prophet and Bagged ARIMA demonstrated greater adaptability and stability. These models were better able to handle sudden shifts in mortality trends and provided the most reliable forecasts across the 12-month horizon. Overall, the findings confirm that effective epidemic forecasting requires flexible models, high-quality data, and an understanding of the underlying public-health context.

Conclusion

This project examined the monthly COVID-19 new-deaths data in Australia by preparing the dataset, analyzing its patterns, and evaluating several forecasting models. The visualizations showed clear waves and an overall trend with no stable seasonal pattern. Based on this behavior, different models were tested, including Holt's Linear, the Damped Trend model, ARIMA, Prophet, and a bagged ARIMA ensemble.

The results showed that the simple smoothing models produced high errors, while ARIMA performed better but was still affected by the sudden changes in the data. Prophet and the bagged ARIMA model achieved the most accurate and stable forecasts, especially when using MAE and RMSE instead of MAPE due to zero-death months. Overall, the findings highlight how flexible and ensemble-based methods provide stronger performance for epidemic time-series forecasting.

References

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Appendices

Appendix A: Data Loading and Handling

```r
# ------------------Start of Fisrt Q1--------------------
# Load required packages
library(tidyverse)
library(tsibble)
library(lubridate)
#--------------------
#DownLoad Data
covid <- read_csv("owid-covid-data.csv")
#--------------------
# Filter the dataset to keep only rows for Australia
# Then select only the columns needed for the analysis (location, date, new_deaths)
aus <- covid %>%
  filter(location == "Australia") %>%      # Keep Australia only
  select(location, date, new_deaths)       # Keep only relevant columns
#-------------------
# Add year-month column for monthly aggregation
aus <- aus %>%
  mutate(month = yearmonth(date))
#------------------
# Handle missing values before aggregation and handing it
aus <- aus %>%
  mutate(new_deaths = if_else(is.na(new_deaths), 0, new_deaths))
sum(is.na(aus$new_deaths))
#-----------------
# Aggregate monthly new deaths
monthly_data <- aus %>%
  group_by(location, month) %>%
  summarize(
    new_deaths = sum(new_deaths, na.rm = TRUE)
  ) %>%
  ungroup()
#-----------------
# Convert the monthly data into a tsibble object
monthly_tsibble <- monthly_data %>%
  as_tsibble(index = month, key = location)
#-------------------
monthly_tsibble %>% head(10)
#--------------------------------End of Q1------------------------
```

Appendix B: Data Visualization

```r
# ------------------ Chapter 2: Data Visualization --------------------
# 2.1 Time Series Plot (Trend)
monthly_tsibble %>%
  autoplot(new_deaths) +
  labs(
    title = "Figure 2.1: Monthly New Deaths from COVID-19 in Australia",
    x = "Month",
    y = "New Deaths"
  )


# 2.2 Seasonal Plot (Seasonality)
monthly_tsibble %>%
  gg_season(new_deaths, labels = "both") +
  labs(
    title = "Figure 2.2: Seasonal Plot of Monthly New Deaths in Australia",
    y = "New Deaths"
  )1

# 2.3 Subseries Plot (Monthly Behavior Across Years)
monthly_tsibble %>%
  gg_subseries(new_deaths) +
  labs(
    title = "Subseries Plot of Monthly New COVID-19 Deaths by Month",
    x = "Month",
    y = "Number of New Deaths"
  )

# 2.4 Autocorrelation Plot (ACF)
monthly_tsibble %>%
  ACF(new_deaths) %>%
  autoplot() +
  labs(
    title = "Autocorrelation Function of Monthly New COVID-19 Deaths",
    x = "Lag (months)",
    y = "ACF"
  )

# 2.5 STL Decomposition
monthly_tsibble %>%
  model(STL(new_deaths ~ trend(window = 13) + season(window = "periodic"))) %>%
  components() %>%
  autoplot() +
  labs(
```

Appendix C: Modeling Estimation Code

```
library(tidyverse)
library(parsnip)
library(tsibble)
library(fable)
library(fabletools)
library(feasts)
library(cli)
library(rlang)
library(modeltime)
library(timetk)
# Split into training and testing data
training_data <- monthly_tsibble %>% filter(month < yearmonth("2022 Aug"))
testing_data <- monthly_tsibble %>% filter(month >= yearmonth("2022 Aug"))
# Modeling (3 models)
data_fit <- training_data %>%
  model(
    Holts_Linear = ETS(new_deaths ~ error("A") + trend("A")  + season("N")),
    Damped_Trend = ETS(new_deaths ~ error("A") + trend("Ad") + season("N")),
    Nonseasonal_ARIMA = ARIMA(new_deaths ~ pdq() + PDQ(0,0,0))
  )
# Modeling for Prophet
prophet_data <- training_data %>%
  rename(ds = month, y = new_deaths)
# Prophet Model
library(prophet)
Prophet_Model <- prophet(prophet_data)

data_fit
Prophet_Model
#-------------------------End of chapter 4----------------------------
```

Appendix D: Model evaluation

```r
# ============================================================
# 5.1 HOLD-OUT TESTING PERFORMANCE
# ============================================================

data_fc <- data_fit %>%
  forecast(h = nrow(testing_data))

model_accuracy <- data_fc %>%
  accuracy(testing_data) %>%
  select(.model, ME, MAE, RMSE, MAPE) %>%
  as_tibble()

model_accuracy

future_prophet <- make_future_dataframe(
  Prophet_Model,
  periods = nrow(testing_data),
  freq = "month"
)

prophet_pred <- predict(Prophet_Model, future_prophet)

prophet_test_fc <- prophet_pred %>%
  mutate(month = yearmonth(ds)) %>%
  filter(month %in% testing_data$month) %>%
  select(.fitted = yhat, month)
```

159:1 (Untitled)

Console | Background Jobs

R 4.5.2 · C:/Users/Lubna/Desktop/.Rproj.user/

```
> library(dplyr)
> library(tidyr)
> library(prophet)
> library(forecast)
> library(tibble)
> library(tsibble)
>
> # ============================================================
> # 5.1 HOLD-OUT TESTING PERFORMANCE
> # ============================================================
>
> data_fc <- data_fit %>%
+   forecast(h = nrow(testing_data))
>
> model_accuracy <- data_fc %>%
+   accuracy(testing_data) %>%
+   select(.model, ME, MAE, RMSE, MAPE) %>%
+   as_tibble()
>
> model_accuracy
# A tibble: 3 × 5
  .model               ME   MAE  RMSE  MAPE
  <chr>             <dbl> <dbl> <dbl> <dbl>
1 Damped_Trend      -2535. 2535. 2684.   Inf
2 Holts_Linear      -4145. 4145. 4549.   Inf
3 Nonseasonal_ARIMA -2201. 2201. 2368.   Inf
>
> future_prophet <- make_future_dataframe(
+   Prophet_Model,
+   periods = nrow(testing_data),
+   freq = "month"
+ )
>
> prophet_pred <- predict(Prophet_Model, future_prophet)
>
> prophet_test_fc <- prophet_pred %>%
+   mutate(month = yearmonth(ds)) %>%
+   filter(month %in% testing_data$month) %>%
+   select(.fitted = yhat, month)
```

```r
prophet_acc_summary <- testing_data %>%
  left_join(prophet_test_fc, by = "month") %>%
  as_tibble() %>%
  select(new_deaths, .fitted) %>%
  mutate(error = new_deaths - .fitted) %>%
  summarise(
    ME   = mean(error, na.rm = TRUE),
    MAE  = mean(abs(error), na.rm = TRUE),
    RMSE = sqrt(mean(error^2,  na.rm = TRUE)),
    MAPE = mean(abs(error / new_deaths), na.rm = TRUE) * 100
  ) %>%
  mutate(.model = "Prophet", .before = 1) %>%
  select(.model, ME, MAE, RMSE, MAPE)

prophet_acc_summary

full_accuracy <- bind_rows(
  model_accuracy,
  prophet_acc_summary
)

full_accuracy
```

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```
> prophet_acc_summary <- testing_data %>%
+   left_join(prophet_test_fc, by = "month") %>%
+   as_tibble() %>%
+   select(new_deaths, .fitted) %>%
+   mutate(error = new_deaths - .fitted) %>%
+   summarise(
+     ME   = mean(error, na.rm = TRUE),
+     MAE  = mean(abs(error), na.rm = TRUE),
+     RMSE = sqrt(mean(error^2,  na.rm = TRUE)),
+     MAPE = mean(abs(error / new_deaths), na.rm = TRUE) * 100
+   ) %>%
+   mutate(.model = "Prophet", .before = 1) %>%
+   select(.model, ME, MAE, RMSE, MAPE)
>
> prophet_acc_summary
# A tibble: 1 × 5
  .model      ME   MAE  RMSE  MAPE
  <chr>    <dbl> <dbl> <dbl> <dbl>
1 Prophet -1449. 1589. 1847.   Inf
>
> full_accuracy <- bind_rows(
+   model_accuracy,
+   prophet_acc_summary
+ )
>
> full_accuracy
# A tibble: 4 × 5
  .model                ME   MAE  RMSE  MAPE
  <chr>              <dbl> <dbl> <dbl> <dbl>
1 Damped_Trend      -2535. 2535. 2684.   Inf
2 Holts_Linear      -4145. 4145. 4549.   Inf
3 Nonseasonal_ARIMA -2201. 2201. 2368.   Inf
4 Prophet           -1449. 1589. 1847.   Inf
>
```

Appendix E: Cross-Validation

```r
full_accuracy

# ============================================================
# 5.2 TIME SERIES CROSS-VALIDATION
# ============================================================

cv_results <- training_data %>%
  stretch_tsibble(.init = 24, .step = 1) %>%
  model(
    Holts_Linear      = ETS(new_deaths ~ error("A")  + trend("A")  + season("N")),
    Damped_Trend      = ETS(new_deaths ~ error("A")  + trend("Ad") + season("N")),
    Nonseasonal_ARIMA = ARIMA(new_deaths ~ pdq() + PDQ(0,0,0))
  ) %>%
  forecast(h = 1)

cv_accuracy <- cv_results %>%
  accuracy(training_data) %>%
  select(.model, ME, MAE, RMSE, MAPE) %>%
  as_tibble()

cv_accuracy
```

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```
Prophet            -1449.  1589.  1847.   Inf
>
> # ============================================================
> # 5.2 TIME SERIES CROSS-VALIDATION
> # ============================================================
>
> cv_results <- training_data %>%
+   stretch_tsibble(.init = 24, .step = 1) %>%
+   model(
+     Holts_Linear      = ETS(new_deaths ~ error("A")  + trend("A")  + season("N")),
+     Damped_Trend      = ETS(new_deaths ~ error("A")  + trend("Ad") + season("N")),
+     Nonseasonal_ARIMA = ARIMA(new_deaths ~ pdq() + PDQ(0,0,0))
+   ) %>%
+   forecast(h = 1)
>
> cv_accuracy <- cv_results %>%
+   accuracy(training_data) %>%
+   select(.model, ME, MAE, RMSE, MAPE) %>%
+   as_tibble()
```

Appendix F: Bootsrapping+Bagging

```r
cv_accuracy

# ============================================================
# 5.3 BOOTSTRAPPING + BAGGING
# ============================================================

set.seed(123)

y_train <- training_data$new_deaths
h <- nrow(testing_data)
n_boot <- 30

boot_matrix <- matrix(NA, nrow = h, ncol = n_boot)

for (i in 1:n_boot) {
  boot_sample <- sample(y_train, replace = TRUE)
  fit_boot <- auto.arima(boot_sample)
  boot_matrix[, i] <- forecast(fit_boot, h = h)$mean
}

bagged_forecast <- rowMeans(boot_matrix)

bagged_acc <- tibble(
  .model = "Bagged_ARIMA_Bootstrap",
  ME   = mean(bagged_forecast - testing_data$new_deaths),
  MAE  = mean(abs(bagged_forecast - testing_data$new_deaths)),
  RMSE = sqrt(mean((bagged_forecast - testing_data$new_deaths)^2)),
  MAPE = mean(abs((bagged_forecast - testing_data$new_deaths) /
                  testing_data$new_deaths)) * 100
)

bagged_acc
```

```
> cv_accuracy
# A tibble: 3 × 5
  .model               ME   MAE  RMSE  MAPE
  <chr>             <dbl> <dbl> <dbl> <dbl>
1 Damped_Trend       334.  724.  925.  57.5
2 Holts_Linear       366.  731.  939.  58.0
3 Nonseasonal_ARIMA  874.  874. 1088.  53.5
>
> # ============================================================
> # 5.3 BOOTSTRAPPING + BAGGING
> # ============================================================
>
> set.seed(123)
>
> y_train <- training_data$new_deaths
> h <- nrow(testing_data)
> n_boot <- 30
>
> boot_matrix <- matrix(NA, nrow = h, ncol = n_boot)
>
> for (i in 1:n_boot) {
+   boot_sample <- sample(y_train, replace = TRUE)
+   fit_boot <- auto.arima(boot_sample)
+   boot_matrix[, i] <- forecast(fit_boot, h = h)$mean
+ }
>
> bagged_forecast <- rowMeans(boot_matrix)
>
> bagged_acc <- tibble(
+   .model = "Bagged_ARIMA_Bootstrap",
+   ME   = mean(bagged_forecast - testing_data$new_deaths),
+   MAE  = mean(abs(bagged_forecast - testing_data$new_deaths)),
+   RMSE = sqrt(mean((bagged_forecast - testing_data$new_deaths)^2)),
+   MAPE = mean(abs((bagged_forecast - testing_data$new_deaths) /
+                   testing_data$new_deaths)) * 100
+ )
>
> bagged_acc
# A tibble: 1 × 5
  .model                  ME   MAE  RMSE  MAPE
  <chr>                <dbl> <dbl> <dbl> <dbl>
1 Bagged_ARIMA_Bootstrap -29.2  361.  465.   Inf
>
```

```r
# ==============================================================
# FINAL TABLE
# ==============================================================

all_accuracy <- bind_rows(
  full_accuracy,
  bagged_acc
)

all_accuracy

```

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```r
# ==============================================================
# FINAL TABLE
# ==============================================================

all_accuracy <- bind_rows(
  full_accuracy,
  bagged_acc
)

all_accuracy
A tibble: 5 x 5
.model                      ME   MAE  RMSE  MAPE
<chr>                    <dbl> <dbl> <dbl> <dbl>
Damped_Trend            -2535.  2535. 2684.   Inf
Holts_Linear            -4145.  4145. 4549.   Inf
Nonseasonal_ARIMA       -2201.  2201. 2368.   Inf
Prophet                 -1449.  1589. 1847.   Inf
Bagged_ARIMA_Bootstrap    -29.2  361.  465.   Inf
```